

REACH evaluation tracks

Five complementary tracks separate ranking, stress, null, natural-prevalence, and label-noise behaviour.

Track A	Controlled real spike-ins • P_HC + B_HC at fixed prevalence 8 datasets × T1-T4 × 5 reps → 160 units (primary)	T1 5-10%	T2 1-5%	T3 0.1-1%	T4 0.01-0.1%
Track B	Splatter synthetic stress • realism-audited 8 datasets × T1-T4 × 3-4 reps → 120 units (secondary)	T1 5-10%	T2 1-5%	T3 0.1-1%	T4 0.01-0.1%
Track C	Null background-only controls • no positives 8 datasets × 4 size tiers × 5 reps → 160 units (false-positive)	T1 5-10%	T2 1-5%	T3 0.1-1%	T4 0.01-0.1%
Track D	Natural blood / CTC prevalence mm_ledergor + breast_ctc_szczerba → 30 units (natural)	natural prevalence			
Track E	Label-noise robustness on Track A expression 4 noise conditions × 160 units → 640 method-units (supervised only)	noisy labels			

Total: 1,110 units per method • ranked Track A subset = 140 (common)